

Report

1. Introduction

This project analyzes synthetic retail transactional data to identify **profit-draining categories, understand inventory turnover, and examine seasonal product behaviour**. SQL, Python (Pandas), and dashboard visualization are used to convert transaction-level data into actionable business insights.

2. Abstract

The project evaluates **1,500 retail transactions from 2024–2025** across different regions, categories, sub-categories, and seasons. Key measures include **revenue, cost, profit, profit margin, quantity sold, stock quantity, and inventory days**.

The analysis shows approximately **₹10.38M revenue, ₹2.36M profit, and a 22.7% overall profit margin**. Beauty has the highest profit margin, while Electronics requires attention because of relatively low margins and high inventory days. The correlation between **Inventory Days and Profit is approximately 0.18**, indicating a weak positive relationship.

3. Tools Used

- **SQL:** Used for data cleaning and analysis of category-wise and sub-category-wise profitability, slow-moving products, overstocked inventory, and seasonal behaviour.
- **Python:** Used to analyze the correlation between inventory days and profitability and seasonal analysis.
- **Power BI:** Used to create an interactive dashboard displaying KPIs, category performance, inventory trends, regional performance, and seasonal behaviour,

4. Steps Involved in Building the Project

1. **Data Import & Cleaning:** Imported the transactional retail dataset into SQL and handled missing/null values.
2. **Profitability Analysis:** Calculated revenue, cost, profit, and profit margins across categories and sub-categories.

3. **Inventory Analysis:** Analyzed inventory days and identified slow-moving and potentially overstocked products.
4. **Correlation Analysis:** Used Python with Pandas to determine the relationship between inventory days and profit.
5. **Seasonal Analysis:** Compared product and business performance across Winter, Summer, Monsoon, and Festive seasons.
6. **Dashboard Development:** Created a dashboard to visualize profitability, inventory performance, seasonal trends, and key business KPIs.
7. **Strategic Recommendations:** Developed recommendations for reducing excess inventory, improving inventory turnover, and focusing on high-margin products.

5. Key Findings

- **Beauty:** Highest profit margin of approximately **39%**, indicating strong growth potential.
- **Home & Kitchen:** Strong profit contributor with approximately **₹7.22 lakh profit** and **23.7% margin**.
- **Electronics:** Main concern, with approximately **44.6 average inventory days** and only **13.1% margin**.
- **Seasonality: Monsoon** generates the highest revenue and profit, while **Summer** has the highest seasonal margin of approximately **23.3%**.
- **Inventory:** Bluetooth Speakers, Headphones, Smartwatches, and Power Banks require attention because of high inventory days and relatively low margins.

6. Conclusion

The project demonstrates how retail transaction data can support practical business decisions. Although the business is profitable overall, category-level differences reveal inventory and margin risks. **Electronics requires tighter inventory control**, while **Beauty offers an attractive high-margin growth opportunity**.

Combining **inventory days, profit margin, sales volume, stock levels, and seasonality** can help improve inventory turnover, reduce overstocking, and strengthen overall profitability